Open Knowledge for a Sustainable Future: Research, Ethics, and Wikipedia

Week 7 — Peer review: how it works

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Roadmap

1. Feedback and Revision
2. Peer Review, Trust, and Metrics
3. Problems: Bias & Inequality, (un)Reproducibility

Writing is a process — feedback is essential

- First drafts are rarely clear, complete, or well-argued
- Feedback (from peers, tutors, self-reflection) helps identify:
  - unclear claims
  - logical gaps
  - missing evidence
  - stylistic issues
- Improving writing increases clarity, persuasiveness, and ethical responsibility — especially important when dealing with data, culture, or community topics

What feedback gives to the writer

- Another reader's perspective — reveals assumptions or blind spots
- Fresh eyes on structure, coherence, and flow: does argument build logically?
- Opportunity to strengthen evidence and address counter-arguments
- Chance to improve style, tone, and accessibility depending on audience

Feedback benefits the critical reader too

- Readers become more critical and informed — able to assess evidence, question assumptions, and spot weak arguments
- Improves collective standards: better writing leads to better discourse
- Feeds back into research culture: clarity and transparency help others replicate, critique, or build further work

Feedback is ethical, not just cosmetic

- Writing about people, cultures, data — carries responsibility. Feedback helps catch problematic framing, bias, or unintended harm
- Encourages accountability: facts must be checked, sources verified, sensitive topics handled with care
- Discipline-wide benefit: careful writing builds trust, respect, and legitimacy for scholarship

Roadmap

1. Feedback and Revision

2. Peer Review, Trust, and Metrics
   - Feedback Beyond Academia
   - Reviewing at the Association for Computational Linguistics

3. Problems: Bias & Inequality, (un)Reproducibility

What peer review is

- Peer review is the process by which experts evaluate a manuscript before publication
- Its aims are to:
  - Check the **soundness** of methods and argument
  - Improve clarity, structure, and contribution
  - Build **trust** in published research
- Reviewers act as gatekeepers and mentors — ideally improving the quality of science and scholarship

Peer review is not about perfection; it is about accountability and conversation.

Types of peer review

- There are several models of peer review, each balancing **transparency, fairness, and accountability** in different ways

Single-blind: Reviewers know the author's identity, but authors do not know the reviewers

- Common in science and social science
- Advantage: reviewers can assess expertise and potential conflicts
- Risk: unconscious bias (e.g., by gender, affiliation, language, or region)

Double-blind: Neither authors nor reviewers know each other's identity

- Aims to reduce bias and level the playing field
- Often used in humanities and linguistics conferences
- Risk: anonymity may limit accountability; reviewers can sometimes guess authors

Open review: Both sides know identities, and sometimes reviews are published

- Promotes transparency and recognition of reviewing labour
- Encourages constructive tone, but may discourage strong criticism
- Used in some open-science and data journals

Peer review: strengths and weaknesses

Strengths

- Enhances rigour and precision
- Provides expert feedback before publication
- Establishes a shared standard of credibility
- Encourages dialogue within a discipline

Weaknesses

- Can be slow, inconsistent, or biased
- Overburdens a small reviewer pool
- Replication and transparency not always checked
- Negative or null results often unpublished (“file-drawer” effect)

Peer review is essential but imperfect — a social process as much as a scientific one.

Evolving models of peer review

- **Post-publication review:**
  - Articles or datasets are published first, then openly discussed
  - Examples: F1000Research, PubPeer
- **Community review:**
  - Large-scale collaborative commenting before publication (e.g., open preprints, GitHub issues, Open Review at LangSci Press)
  - Builds transparency and collective expertise
- **Preprint feedback:**
  - Early sharing (e.g., arXiv, LingBuzz) invites informal peer feedback before formal submission
  - Speeds dissemination and reduces “scoop” anxiety
- These models shift peer review from an evaluation to a conversation

Ethical responsibilities of peer reviewers

- Reviewers are entrusted with **privileged access** to unpublished work
- Ethical responsibilities include:
  - **Confidentiality**: do not share, quote, or use ideas from the manuscript
  - **Impartiality**: declare conflicts of interest; avoid personal or institutional bias
  - **Constructive criticism**: focus on improving the work, not attacking the author
  - **Timeliness**: return reviews promptly; long delays can harm authors' careers
  - **Accuracy**: evaluate methods, evidence, logic, and citations fairly
  - **Respect**: use professional tone; assume good intent; avoid dismissive language
- As Rockwell (2005) emphasises, access to unpublished research creates a **power imbalance**: reviewers must not exploit this advantage

A good reviewer protects confidentiality, improves clarity, and strengthens trust in scholarship.

Feedback in industry: the practice of “giving notes”

- In many industries (film, publishing, games, advertising), feedback is formalised as **“notes”** (Catmull & Wallace, 2014)
- Notes come from editors, producers, managers, clients — often in a hierarchical chain
- The goal is similar to academic review:
  - improve clarity, impact, and quality
  - ensure the work meets its purpose and audience
- But constraints differ:
  - strict deadlines, budgets, brand guidelines, legal concerns
  - competing visions between writer/creator, editor, and client

Video: Ed Catmull on the Pixar Braintrust (Stanford, 8 min)

How industry notes differ from academic peer review

- **Hierarchical:** notes often come from someone with authority over the project (editor, producer, manager)
- **Goal-driven:** feedback must serve the needs of the project, brand, or product — not the advancement of knowledge
- **Collaborative but constrained:** creators negotiate changes, but ultimate decisions may be external
- **Iterative cycles:** multiple rounds of notes → revisions → more notes → final approval
- **Success = alignment** with project goals, not independent scholarly rigour

Best practices from industry feedback culture

- **Clarity first:** Notes should define the problem before proposing solutions (Pixar rule)
- **Respect creative intent:** Improve the work without imposing a personal style
- **Prioritise high-impact changes:** Not all notes can be addressed; choose the ones that serve the project
- **Negotiate professionally:** Creators can ask for clarification and push back when necessary
- **Shared responsibility:** Good notes identify issues; creators choose how to solve them

What students can learn from industry feedback

- You will encounter feedback in many forms: editorial, managerial, client-oriented, or technical
- Key transferable skills:
  - receiving feedback without defensiveness
  - interpreting conflicting goals
  - deciding which suggestions to adopt, adapt, or reject
  - communicating clearly about revisions
- Ethical themes remain:
  - respect, clarity, collaboration, responsibility

Good feedback — academic or industrial — improves the work and strengthens the team.

Feedback cultures beyond academic peer review

- **Journalism:** Editors and fact-checkers verify claims against primary sources before publication — different from academic review, which often assumes sources are cited correctly
- **Wikipedia:** Articles pass through multiple stages of community review; good/featured article criteria require verifiability and neutral point of view
- **Code review:** In software development, peers review code for bugs, style, and maintainability — feedback is typically public and tied to version control (e.g., GitHub pull requests)

Feedback aims to improve quality through structured critique before the results are made public

Peer review in Computational Linguistics

- Peer review is so central to scholarship that whole communities invest in improving it
- The Association for Computational Linguistics (ACL) publishes:
  - reviewer training guides (ACL 2023 Program Committee, 2023)
  - best-practice recommendations (ACL Rolling Review, 2025)
  - annual reports analysing what works and what fails (ACL Standing Committee on Peer Review, 2025)
- These offer a model for responsible, constructive, ethical reviewing
- A large conference at scale — ACL 2025, for example:
  - 8,360 submissions
  - 1,699 accepted to the main conference (20.3%), 1,392 as Findings (16.7%)
  - 169 Senior Area Chairs, 1,937 Area Chairs, 11,720 reviewers!
  - Each paper reviewed by three reviewers, the authors respond, the reviewers update their reviews, the area chairs suggest which papers are accepted, the senior area chairs make the final decisions, …

Accept reviews only when appropriate (ACL guidelines)

- Before agreeing to review, check:
  - **Expertise**: Can you fairly assess the methods and claims?
  - **Conflicts of interest**: shared institution, collaborators, competitors
  - **Capacity**: Do you actually have time to read carefully?
- Accepting when you are unqualified or overloaded leads to unfair or rushed reviews
- Slightly undermined by the fact that if you submit a paper, someone from the author team must volunteer to review, and owes four reviews per submission (capped at 16 in a cycle), …
  - ACL 2025 accepted 20.3% to the main conference and a further 16.7% as Findings — four papers miss the main conference for every one that makes it, and nearly two thirds are turned down altogether
  - From October 2026 the ACL sustainable reviewing policy goes further: each submission must be backed by a qualified service contributor (who may serve for at most two submissions), authors are capped at 20 submissions and 5 first-author submissions a cycle, and papers whose authors do not register and do the service are desk rejected

Read carefully and generously (ACL guidelines)

- First read: understand the research question and contribution
- Second read: evaluate evidence, structure, and clarity
- Look for what the author is *trying* to do
- Avoid snap judgments based on style, formatting, or first impressions

Write constructive, actionable feedback (ACL guidelines)

- Every review should include both:
  - **Strengths** — what works well
  - **Weaknesses** — what needs improvement
- Make criticisms specific:
  - point to a sentence or section
  - explain why it is unclear
  - suggest how to fix it
- Vague comments (*paper is confusing*) are unhelpful

Check evidence, ethics, and reproducibility (ACL guidelines)

- A reviewer is responsible for more than novelty
- Consider:
  - Are claims supported by evidence?
  - Are methods and data described clearly?
  - Are ethical implications acknowledged?
  - Can results be verified or reproduced?
- Responsible peer review strengthens the entire field

Professionalism and respect (ACL guidelines)

- Use a constructive and collegial tone
- Avoid dismissive language:
  - “This is trivial”
  - “obviously wrong”
  - “the authors don’t understand”
- Remember: reviewers and authors are colleagues
- Treat the paper the way you would want yours to be treated

Reviewer power and responsibility (ACL COPR)

- Reviewers gain **access to unpublished work** before the community
- This creates responsibility:
  - Never use unpublished ideas in your own work
  - Never delay a review to gain priority
  - Never punish papers that challenge your favourite theory
- Peer review is a trust-based system; unethical reviewing harms entire fields

What this means for you (student reviewers)

- The same principles apply in the classroom:
  - read generously
  - comment specifically
  - focus on clarity, evidence, and logic
  - be respectful and supportive
- Practising good peer review improves your own writing:
  - you learn what makes an argument work
  - you see how readers interpret drafts
  - you build editorial judgment

Based on ACL reviewer training materials (ACL 2023 Program Committee, 2023; ACL Rolling Review, 2025; ACL Standing Committee on Peer Review, 2025).

ACL-inspired peer review checklist

- Am I qualified and conflict-free?
- Have I read the paper carefully (twice)?
- Did I identify strengths and weaknesses?
- Are my comments specific and actionable?
- Did I evaluate ethics, evidence, and reproducibility?
- Is my tone respectful and professional?
- Have I upheld confidentiality and integrity?

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   - Punktoza
   - Bias and Inequality in Peer Review
   - The Reproducibility Crisis
   - AI and Peer Review

Publish or Perish

- Academics are under pressure to publish
- This is a major part of how they are evaluated
- This should lead to more, higher quality publications
- But can lead to many more, lower quality publications
  - Or even fraud!
- And means we have to review more and more publications

Metrics: why they were introduced

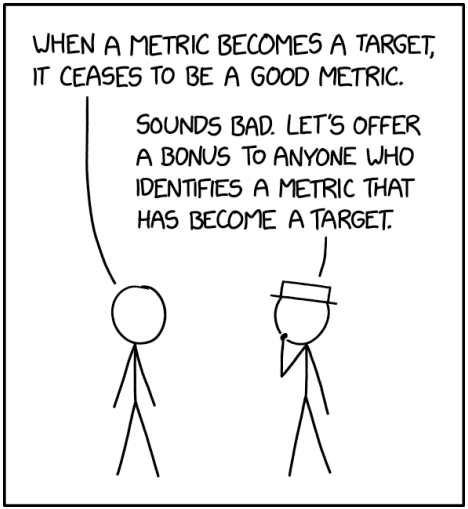
- Growing research ecosystems needed **standardised indicators** to compare:
  - journals (Impact Factor)
  - researchers (h-index)
  - institutions (citation counts, productivity)
- **Impact Factor (IF):** citations made in one year to what a journal published in the *two preceding* years, divided by the number of "citable items" (research articles and reviews) it published in those two years
  - Designed to help libraries choose journals and to signal visibility
  - A standard criticism is the mismatch between the two halves: the numerator counts citations to *everything* (editorials, letters, news), the denominator counts only the "citable items"
- **h-index:** a researcher has index *h* if they have *h* papers each cited at least *h* times
  - Intended to combine productivity + influence in a single number
- Metrics aimed to make evaluation **transparent, comparable, and data-driven**
- They were introduced to help — not to replace — expert judgment

Metrics: when measures become targets

- When hiring, promotion, or funding is tied to metrics:
  - Researchers optimise for what is counted, not what is meaningful
  - Citation counts become currency; novelty outranks replication or rigour
- **Goodhart's Law**, as Strathern (1997, p. 308) famously put it:
  When a measure becomes a target, it ceases to be a good measure.
  - IF and h-index lose validity once people actively game them
- **Punktoza ("points fever"):** publishing strategically to accumulate evaluation points rather than to advance knowledge (Kulczycki, 2017a, 2017b)
  - Encourages salami-slicing, low-risk incremental work, quantity over substance
  - A systemic response to metric-driven national evaluation frameworks
- Metrics intended to guide evaluation end up **shaping behaviour**, sometimes harming scholarship

Quantifying trust can erode it when numbers replace judgment.

Goodhart's Law xkcd 2899



Peer review is not neutral — known biases

- Peer review should be meritocratic — but social biases can shape outcomes
- Empirical studies document systematic bias based on gender, institutional prestige, and affiliation
- This can reproduce inequalities in academic visibility and access

Gender bias and underrepresentation

- A study of 145 journals — 1.7 million authors and 740,000 referees — found that manuscripts by women were not universally penalized, yet women remain under-represented among reviewers and editors (Squazzoni et al., 2021)
- An earlier study of 9,000 editors and 43,000 reviewers at one publisher found the same under-representation, and that reviewers tend to be matched with authors like themselves (Helmer et al., 2017; Murray et al., 2018)
- In grant-review contexts, women have also been shown to receive lower scores despite similar qualifications (Wennerås & Wold, 1997)

Institutional – affiliation and prestige bias

- "Affiliation bias" has been measured directly: WSDM 2017 ran a controlled experiment in which every paper got two *single-blind* reviewers and two *double-blind* ones (Tomkins et al., 2017)
  - The single-blind reviewers were markedly more likely to recommend acceptance for papers by famous authors (odds $\times 1.63$) and from top universities ($\times 1.58$) or top companies ($\times 2.10$)
  - Their double-blind colleagues, reading the same papers, showed no such advantage — so double-blinding takes it away
- This can disadvantage authors from lesser-known institutions, even when the manuscript quality is comparable
- Result: citation networks and publication records amplify institutional inequality — making "the rich institutions" richer

Open review — which is what you just did

- Your reviews went in the Moodle forum: **signed**, and readable by the whole class. That is **open review**, and some journals work this way (eLife, BMJ, F1000).
- **What blinding buys you**: double-blinding removes the famous-author advantage (Tomkins et al., 2017).
- **What signing costs you**: in a randomised trial at the *BMJ*, reviewers asked to be identified to the author were **12 points more likely to refuse** to review at all (35% v 23%) — but for those who did review, quality, recommendation and time taken were **no different** (van Rooyen et al., 1999).

So, from your own week:

- Did you soften anything because your name was on it?
- Did reading other people's reviews change how you wrote yours?
- Would you have said more, or less, anonymously?

Why these biases happen — and what can we do?

- Why these biases happen
  - Reviewers (and editors) may rely on stereotypes, familiarity, or “status cues” (institution name, authors’ reputation) rather than content
  - Reviewing is human work; unconscious bias — on gender, prestige, nationality, career stage — is common
  - Peer-review bias is not just about individual reviewers, but structural patterns: who gets invited to review, whose papers get visibility, who gets accepted
- What this means for us
  - As reviewers: stay alert to bias. Evaluate based on content, not identity or institution
  - As authors: be aware that results and visibility may be influenced by social factors beyond quality
  - As a community: encourage diversity in reviewers/editors, anonymised review where possible, transparency in editorial decisions
  - Use blind review where possible

What is reproducibility / replicability?

- **Reproducibility:** re-analyse the same data/code to get the same result
- **Replicability:** collect new data under same conditions and test whether the effect holds
- Both are fundamental for reliable knowledge — without them, published findings remain provisional (Goodman et al., 2016)

What the crisis looks like — some sobering numbers

- In a 2015 project replicating 100 psychology studies, only about 36% of results were successfully replicated; effect sizes dropped by about 50% (Open Science Collaboration, 2015)
- A 2016 *Nature* survey of 1,576 researchers, across all the sciences, found over 70% had tried and failed to reproduce another scientist's results — and 52% judged that there is a significant crisis (Baker, 2016)
- Cancer biology is no better: an eight-year project repeated 50 experiments from 23 high-profile papers, and only 46% of the replications (51/112) succeeded — with effect sizes a median 85% smaller (Errington et al., 2021)

Why this matters

If published findings routinely fail replication, the scientific literature becomes unreliable — which undermines trust, wastes resources, and misleads follow-up research.

Video: Veritasium: "Is Most Published Research Wrong?" (12 min)

Causes: why reproducibility fails

- Low statistical power (too few subjects/samples) — leads to unstable, non-robust effects (Ioannidis, 2005)
- Publication bias & questionable research practices (QRPs): only positive or novel results tend to get published; null results stay hidden
- Poor documentation, missing code/data, hidden methods — even honest researchers make replication hard when they do not share full materials
- Time & incentive structures: scientists rewarded for novelty, impact, and speed — not for replication or careful method sharing

How FAIR + CARE + Good Writing helps fix reproducibility

- Proper metadata, shared code/data, and clear methods (FAIR) make re-analysis possible
- Ethical responsibility (CARE) encourages transparent, honest reporting — including limitations, context, and reproducibility risks
- Careful writing and documentation help readers understand exactly how a study was done
- Together — transparent data practices, open sharing + honest reporting

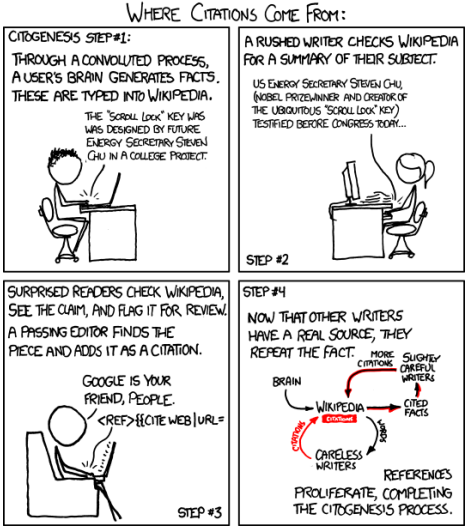
= more reliable science

Citation as trust

- A citation is a **link of trust**: it says “I have read and found this work reliable enough to build upon”
- Over time, citations form **networks of credibility** — maps of intellectual influence
- Not all citations are equal:
  - Some endorse, others critique or merely mention
  - “Citation cascades” can perpetuate unverified claims
- Responsible citation means reading, understanding, and acknowledging context — not just counting links

Citation is conversation: each reference extends a thread of shared trust.

Citogenesis xkcd 978



AI in peer review: a known problem

- Large language models (LLMs) like ChatGPT are increasingly used in academic writing — and in peer review
- A 2024 study estimated that 6.5–17% of peer reviews at major AI conferences may have been substantially written or modified by LLMs (Liang et al., 2024)
- Telltale signs: unusual adjectives (“commendable”, “meticulous”, “intricate”) spiked 10–35× in frequency
- Major publishers all restrict AI in peer review, but not in the same way:
  - **All of them:** never upload the manuscript, or any part of it, to a public AI tool — that alone breaks confidentiality
  - **Taylor & Francis** and **Sage:** do not let AI generate the review at all (T&F allow it only to improve the language of a review you wrote yourself)
  - **Springer Nature:** AI may support a reviewer but never replace their judgement — use only secure or institutionally approved tools, and be transparent about it

Why AI-generated reviews are problematic

- **Confidentiality breach**: uploading a manuscript to an LLM shares it with a third party
- **Accountability**: reviewers are responsible for accuracy and judgment — LLMs cannot be held accountable
- **Bias and hallucination**: LLMs can perpetuate biases and produce plausible-sounding but incorrect critiques
- **Superficiality**: AI-generated reviews tend to be generic and miss domain-specific issues
- **Trust erosion**: if reviews are automated, peer review loses its meaning as expert evaluation

Peer review depends on human expertise, judgment, and responsibility — qualities LLMs cannot replicate.

Typical guidance on AI in peer review

- Most journals and conferences now require:
  - Disclosure of any AI assistance in writing or reviewing
  - Human accountability for all submitted content
  - No uploading of confidential manuscripts to AI tools
- Acceptable uses (with disclosure):
  - Grammar and spell-checking
  - Improving clarity of your own writing
- Unacceptable uses:
  - Generating review text
  - Summarising or analysing manuscripts you are reviewing
  - Submitting AI-generated content as your own expert judgment

Acknowledgements

- I have had over 200 papers reviewed (thanks reviewers!)
- I have peer reviewed over 2,000 conference papers, 100 journal papers, 10 PhDs, 2 Habilitations, and many grants
- OpenAI (2025) was used to format the references, and generate a first draft of the slides
- **Claude Opus 5** (Anthropic, 2026) was used to check the slides: to resolve every DOI against Crossref, fetch every URL, and compare each quotation with its original. The prompt was of the form

  ```
  Verify every claim, DOI, URL and quotation against a real source. Where you cannot
  verify something, say so rather than guessing --- never invent a reference.
  ```

  Each correction names the source it was checked against. Responsibility for what remains is mine.

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